Suction Apparatus



Category: **Medical, Healthcare & First Aid** Type: **Full SOP** Priority: **High** 3 labs

Covers these lab items: Suction Apparatus

What it is used for

A suction apparatus removes blood, mucus, secretions, or other fluids from a patient's airway or a designated clinical site when clinically indicated. Correct setup, appropriate suction settings, infection control, and patient monitoring help provide effective suction while reducing tissue injury and contamination risks.

Main parts

- Vacuum pump generates negative pressure.
- Collection canister collects aspirated fluids.
- Patient tubing connects the suction source to the collection system.
- Suction catheter or approved clinical attachment removes fluids from the intended site.
- Pressure regulator adjusts suction strength.
- Vacuum gauge indicates the available negative pressure.
- Bacterial filter helps protect the pump where fitted.
- Power switch, battery, or foot control operates the unit depending on the model.

Before you start

1. Only trained healthcare personnel should operate the apparatus within their clinical scope and local protocols.
2. Confirm the intended clinical use and select the correct suction device and approved accessories.
3. Inspect the housing, power cable, tubing, canister, seals, filter, and connectors for damage or contamination.
4. Check that the collection canister is correctly fitted and below its maximum fill level.
5. Verify the power supply or battery condition and test the vacuum gauge and controls.
6. Set the prescribed suction level according to the patient, procedure, and device instructions.
7. Perform hand hygiene and wear appropriate PPE; use sterile accessories when required by the procedure.

How to use

1. Place the apparatus on a stable surface near the patient without obstructing emergency access.
2. Connect the collection canister, tubing, filter, and approved patient attachment according to the device diagram.
3. Confirm that all connections are secure and the canister lid seals correctly.
4. Switch on the apparatus and occlude the patient end briefly as instructed to check vacuum function.

5. Adjust the pressure regulator to the prescribed setting and verify the gauge reading.
6. Explain the procedure to the patient when feasible and position the patient appropriately for the clinical indication.
7. Insert and use the suction catheter only according to the relevant clinical protocol and training.
8. Apply suction for the prescribed duration while observing the patient and maintaining appropriate technique.
9. Monitor oxygenation, breathing, heart rate, comfort, and other clinically relevant signs throughout the procedure.
10. Stop suction immediately if significant distress, bleeding, abnormal response, or equipment failure occurs.
11. After use, switch off the unit, disconnect accessories safely, and dispose of single-use items in designated waste streams.
12. Clean and disinfect reusable parts according to the manufacturer's instructions and document the procedure as required.

⚠ Safety

- Never use suction settings or durations beyond the prescribed clinical limits or device instructions.
- Use the correct catheter size and technique to reduce mucosal injury and other complications.
- Do not allow the collection canister to overfill or fluid to enter the pump.
- Use appropriate infection-control precautions and never reuse single-use accessories.
- Do not use damaged electrical components or operate the device in an unsuitable wet environment.
- If suction is needed for an airway emergency, follow the applicable emergency protocol and obtain appropriate clinical assistance.

Care & storage

- Empty and dispose of collected fluids according to clinical infection-control and waste procedures.
- Replace single-use canisters, filters, tubing, and catheters as specified by the manufacturer and facility policy.
- Clean and disinfect reusable components after each use using approved products.
- Inspect seals, tubing, connectors, and the vacuum gauge regularly.
- Arrange preventive maintenance and remove the device from service if suction performance is unreliable.

Common mistakes to avoid

- Setting suction pressure without considering the patient and clinical indication.
- Allowing the collection canister to overfill or the filter to become wet.
- Reusing disposable accessories or neglecting infection-control procedures.
- Failing to monitor the patient or verify suction performance before use.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.